FWA-05 / HEALTHCARE FWA KNOWLEDGE BASE

Dental FWA

Service consistency, bundling and staged treatment review

Synthetic dental payment policy

Scope

This fictional dental knowledge base supports a RAG exercise about claims consistency. It contains no real DOH data, actual tariff or official coding rule. Procedure labels D-SYN-EXAM, D-SYN-IMAGE, D-SYN-RESTORE and D-SYN-BUNDLE are invented local codes. They must not be substituted for any real dental coding system.

DE-P01 | Clinical and service identifiers

A line must include a valid synthetic member, provider, service date, local procedure label, tooth identifier when applicable, surface when applicable and service status. A missing tooth identifier is a documentation gap for tooth-specific services; it is not automatically proof of a false service. Use a documented local tooth schema and validate range without guessing missing values.

DE-P02 | Bundle rule

In this exercise, D-SYN-BUNDLE includes the associated D-SYN-EXAM performed by the same provider for the same member on the same date. Separate payment for both is a bundle-conflict candidate. A separately documented unrelated emergency visit may qualify for an exception after review. This invented exception requires evidence; it is not inferred merely from a modifier.

DE-P03 | Repeated service and staged care

Repeated tooth-specific restoration within 60 days triggers review, not automatic denial. Review tooth and surface, initial procedure status, replacement linkage and staged treatment notes. A planned multi-stage procedure may legitimately generate multiple records. A corrected claim replaces its linked original for active payment analysis.

Decision control

A trained reviewer assesses documentation and the fictional contract configuration. The RAG assistant should explain a policy and its evidence requirements without deciding whether a real dental treatment was clinically necessary. Keep financial integrity questions distinct from patient-specific clinical judgments.

Workflow and illustrative metric definitions

Validate and match

Normalize local tooth and surface fields; keep original text. Reconcile replacement claims and voids. Match potential conflicts using member_id, provider_id and service_date; use tooth and surface for relevant repeat-service rules. Do not match records solely on a procedure label when the services could relate to different teeth.

Review packet

Include both conflicting lines, treatment timeline, relevant tooth/surface, paid amounts, fictional policy version and supporting notes. Check whether an emergency exception or staged treatment is documented. Record supported, corrected, overpayment confirmed, insufficient evidence or referred for specialist review. Escalate documentation uncertainty before asserting intent.

Synthetic portfolio

Measure	Value / definition
Accepted dental lines	3,000
Bundle-conflict lines	90
Repeat-service lines	75
Overlap across rules	15 lines
Unique candidate lines	90 + 75 - 15 = 150
Candidate line rate	150 / 3,000 = 5.0%
Reviewed candidate lines	50
Confirmed adjustment lines	20 / 50 = 40% reviewed yield
Confirmed / recovered SCU	8,000 confirmed; 5,000 verified recovered

Interpretation

The 150 candidates are claim lines, not visits, teeth, patients or proven cases. Count unique encounters separately using encounter_id if available. The recovered share of the confirmed amount is 5,000 / 8,000 = 62.5%; this does not measure model recall. A 40% reviewed yield from a prioritized sample cannot estimate the prevalence of unsupported dental care.

Operational measures

Track missing tooth-field rate, median documentation turnaround, appeal outcomes and rule-specific false-alert reasons. Stratify repeat-service patterns by service category and staged-care status. Compare old and new rules on the same frozen snapshot before attributing a change in alerts to provider behavior.

Dental scenarios and exception handling

DE-S01 | Bundle with no exception

A synthetic member receives D-SYN-BUNDLE and D-SYN-EXAM from the same provider on 12 September. Both are paid and there is no documented separate emergency visit. Expected action: request supporting records and apply DE-P02 after review. If the included exam was separately paid without support, calculate the adjustment from its actual paid amount.

DE-S02 | Different teeth

A member has two D-SYN-RESTORE records within 20 days, one for local tooth T11 and one for T12. A procedure-only matching rule flags them. Expected result: inspect tooth and surface, and close the same-tooth alert if the records establish different services. Improve matching rather than forcing a provider-level adverse conclusion.

DE-S03 | Staged treatment

Two records concern the same tooth within 30 days. The treatment plan identifies a temporary stage followed by definitive treatment. Expected result: specialist review of DE-P03 and the fictional benefit configuration. The mere presence of staged notes is evidence to assess, not a guarantee of separate payment.

DE-S04 | Missing image

The review packet lacks an image. This corpus does not state that every restoration requires imaging. A correct answer identifies the evidence gap and refuses to invent an imaging mandate. If a separate policy is supplied later, index it with its effective date and assess whether it applies to the service.

Evidence and limitations

Tooth notation can differ between systems; a local T11 identifier has no assumed mapping to an international notation in this exercise. An OCR error can change a digit and create an apparent different-tooth service. Capture extraction confidence and compare with the original document before correcting a clinical identifier.

Analytics check

A dashboard may count rules, lines and review cases differently. Label each chart with its unit and denominator. Retain overlap counts so users can reconcile the unique candidate total. Make the policy exception discoverable from the alert detail view rather than hiding it in an unrelated appendix.

Hybrid retrieval and interview practice

Exact-label challenge

Ask: What is included in D-SYN-BUNDLE? A dense retriever may match general dental text but miss the exact invented label. Compare keyword retrieval, vector retrieval and a hybrid method using the same questions. Check that both the inclusion rule and emergency exception are present in the retrieved context.

Hybrid and reranking exercise

Retrieve candidate passages from keyword and vector indexes, combine them with a documented ranking method, remove duplicates and rerank a bounded shortlist. Measure evidence recall before and after reranking. Do not assume that a higher semantic score means the policy applies to the correct date or provider.

Structured lookup exercise

Use a structured query to find claim pairs sharing member, provider, date and tooth. Use RAG to explain the applicable policy and exceptions. Do not ask embeddings to perform exact joins or sum financial values. Store the executed query version and source snapshot for reproducibility.

Interview Q1 | Dense versus lexical retrieval?

Dense retrieval can match meaning across different wording; lexical retrieval is valuable for exact identifiers and uncommon procedure labels. Hybrid retrieval can improve coverage but adds tuning and evaluation cost. Interview Q2: What does reranking do? It reorders retrieved candidates using a more detailed query-passage comparison; it cannot recover a passage never retrieved.

Interview Q3 | How would you handle scanned PDFs?

Detect textless or low-quality pages, apply OCR, retain page provenance and inspect extraction quality. Test tables, numbers and identifiers explicitly. A fluent answer based on corrupted OCR is still wrong. Interview Q4: Why use tools for sums? Deterministic calculations are reproducible and easier to validate than free-form model arithmetic.

Advanced extension

Add query rewriting for synonyms while preserving exact local labels, dates and permissions. Compare answer quality, latency and token cost against a simpler baseline. Introduce the additional retrieval steps only when evaluation shows a useful improvement. See FWA-06 for experiment design and production-readiness gates.